MIL-M-12320A(SigC)
Amendment 4
14 September 1973
SUPERSEDING
Amendment 3
28 February 1964

MILITARY SPECIFICATION

MAST BASES, SECTIONS, AND ADAPTER KIT

This amendment forms a part of MIL-M-12320A(SigC), dated 17 July 1958, and is approved for use by all Departments and Agencies of the Department of Defense.

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1.1: Add the following to the list:

"Mast Base AB-558( )/GR

Mast Base AB-652( )/GR"

2.1: At the end of the first sentence add: "to the extent specified herein:" Under DRAWINGS, SIGNAL CORPS add the following to the list:

"SC-DL-75827 Mast Base AB-558( )/GR

SC-DL-49678 Mast Base AB-652( )/GR"

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2.2: Replace the entire paragraph with the following:

"2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

ASSOCIATION OF AMERICAN RAILROADS

Uniform Freight Classification No. 4 Consolidated Freight Classification Rule No. 21.

FSC 5985

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(Application for copies should be addressed to the Uniform Freight Classification Committee, Tariff Publishing Officer, Room 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

AMERICAN TRUCKING ASSOCIATION

* National Motor Freight Classification No. A-13.

(Application for copies should be addressed to the National Classification Board, 1616 P Street, N. W. Washington 20036, D.C.)."

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3.1: After AB-15( )/GR in first sentence add "AB-558( )/GR, AB-652( )/GR."

3.5.1: Delete "Except as stated in 3.5.1(b)."

3.5.2(b): Replace " -150°F" with " +150°F."

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3.8: Add, "Not applicable to the AB-652( )/GR."

* 3.9: At end add, "Mast Base AB-652( )/GR shall withstand 300 inch-pounds of torque instead of 150 inch-pounds."

3.10.2: Add "Except for the neoprene covering of Mast Base AB-652( )/GR which shall be flexible and capable of bending through 90 degrees while maintained at a temperature of -65°F, without developing any cracks.

3.11: Replace last sentence with the following: "The bending moment, measured from the axis of the mast base, required for a 90 degree bend shall be as follows:

(a) Mast Bases MP-65( ) and MP-76( ): 600 to 800 inch-pounds.

(b) Mast Bases AB-15( )/GR and AB-558( )/GR and Antenna Spring Section AB-129( )/PR: 180 to 260 inch-pounds.

* (c) Mast Base AB-652( )/GR: 700 to 900 inch-pounds."

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Add the following new paragraphs:

"3.11.1 Cold bend recovery.- After the top ferrule section (stem portion) of Mast Base AB-652( )/GR, with attached vertical 15-foot whip, has been deflected to assume an angle of 45-degrees with respect to the vertical for a 24-hour period at specified temperature, the antenna assembly shall recover to the corresponding position listed below, when released.

	Temperature (24-hour period)	Recovery after 45° Deflection
(a)	-60 + 0°F - 5°F	within 12 ± 3 degrees of vertical
(b)	0 ± 5°F	within 10 ± 3 degrees of vertical
(c)	75 ± 10°F	within 4 ± 2 degrees of vertical"

3.13: In line three, replace "Mast Base AB-15( )/GR" with "Mast Bases AB-15( )/GR and AB-558( )/GR." Add the following sentence: "Mast Base AB-652( )/GR shall be capable of withstanding 200-90 degree flexures, followed by 800-75 degree flexures followed by 2,000-45 degree flexures, all flexures at a rate between 20 and 30 per minute."

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Add the following paragraph at the beginning of Section 4:

"4. RESPONSIBILITY FOR INSPECTION.

Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements."

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4.1: Replace with the following:

"4.1 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.2)

(b) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.3)

(2) Quality conformance inspection of preparation for delivery. (See 4.14)"

4.2: Replace with the following:

"4.2 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in Table I, and the inspection specified for Group A, Group B and Group C (see Tables II, III, and IV, respectively)."

Table I shall remain unchanged.

4.3: Replace with the following:

"4.3 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.3 through 4.3.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements."

4.3.2: Replace with the following:

"4.3.2 Group B inspection.- This inspection, including sampling, shall conform to table III and to the special procedures for small sample inspection of Standard MIL-STD-105. The AQL shall be 4.0 percent defective and the special inspection level shall be S-4. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection."

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4.3.2.1: Delete.

4.3.2.2: Delete.

TABLE IV. Group C inspection. Add the following test:

"Cold-Bend Recovery 3.11.1 4.11.1"
(for Mast Base AB-652( )/GR)

4.3.3: Replace with the following:

"4.3.3 Group C inspection. This inspection shall be as listed in Table IV, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection."

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4.3.4: Replace the entire paragraph with the following:

"4.3.4 Reinspection of conforming Group B sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group B inspection may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection."

4.6, 4.6.1, and 4.6.2: Add to each: "Not applicable to Mast Base AB-652( )/GR."

★ 4.7: In the first sentence, after "150 inch-pounds" insert "(300 inch-pounds for Mast Base AB-652( )/GR)."

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4.9: In the first sentence, after " -40°F." insert "(-65°F for Mast Base AB-652( )/GR)."

4.11: Replace the last sentence with the following: "The bending moment, measured from the axis of the mast base, required for a 90-degree bend shall be in accordance with the requirements of 3.11." Add the following new sub-paragraph:

"4.11.1 Cold-Bend recovery test.- A simulated 15-foot whip, a 2½-pound weight securely attached to rigid extension to the mast base so that it is 20.9-inches above the top of the base (the extension can be an MS-116( ) Mast Section or similar rod as approved by the contracting officer), shall be installed on Mast Base AB-652( )/GR. The base shall be firmly secured such that the extension is in a vertical position. The extension shall be deflected and secured so that the top stem portion of the mast base is at an angle of 45 degrees from the original vertical position. The temperature of the assembly shall be lowered to -60°F and maintained for 24 hours. While still in the -60°F environment the top stem portion restraint shall be released and the recovery shall be measured and be in accordance with 3.11.1. Repeat testing at 0°F and room temperature (75 ± 10°F)."

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4.12: Replace with the following:

"4.12 Test for shock-absorbing qualities of mast bases.- With the mast base rigidly secured, the stem portion shall be given blows of such force that the stem is deflected to specified angle. The angle of flexure shall be measured between the bottom stem and the top ferrule socket of the mast base, with all bends being in the same direction. The required number of blows may be accumulated intermittently to prevent excessive heat buildup in the mast base.

(a) Mast Bases MP-65( ) and MP-76( ): 2,500 blows at the rate of one blow per second; flexed through 90 degrees by each blow.

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(b) Mast Bases AB-15( )/GR and AB-558( )/GR and Antenna Spring Section AB-129( )/PR: 1,500 blows at the rate of one blow per second; flexed through 90 degrees by each blow.

(c) Mast Base AB-652( )/GR: 200 blows, 90 degree flexure by each blow; 800 blows, 75 degree flexure by each blow; 2,000 blows, 45 degree flexure by each blow. All flexures shall be at a rate between 20 and 30 per minute and recovery after each flexure shall be gradual simulating recovery of the base supporting a 15-foot whip."

4.14 and paragraphs 4.14.1 through 4.14.2.3, replace with the following:

"4.14 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5."

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* 5.2.3: In line 4, replace "No. A-3 and No. 14" with "No. A-13."

NOTE: The margins of this amendment are marked with an asterisk to indicate where changes from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodian:
Army-EL

Preparing activity:
Army-EL

Project No. 5985-A408

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